What Determines Case?

The Current Study

Method

Results

Discussion

Conclusion

Ger	Eng
an	"at"
auf	"on"
hinter	"behind"
in	"in"
neben	"beside"
über	"above"
unter	"under"
vor	"before"
zwischen	"between"

The Case of German Two-Way Prepositions

Daniel Jach, PhD

Southwest Jiaotong University

July 3, 2021 – online – LxGr2021

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A Peculiar Case

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A Peculiar Case

(1)

Najara	ging	zu	dem	alten	Mann	hinüber.
PR	went	to	the-DAT	old	man	over

"Najara went to the old man."

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A Peculiar Case

(1) Najara ging zu dem alten Mann hinüber.
PR went to the-DAT old man over

"Najara went to the old man."

(2) Der Wind rauschte um das-ACC Haus
the wind rushed around the-ACC house

"The wind rushed around the house."

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(2)

Der	Wind	rauschte	um	das-ACC	Haus
the	wind	rushed	around	the-ACC	house

"The wind rushed around the house."

(3) a.

Susi	reist	in	die	Heimatländer	der	Schulkinder.
PR	travels	in	the-ACC	homelands	of the	school children

"Susi travels into the homelands of the school children."

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b. Susi reist in den Heimatländern der Schulkinder.
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Overview

1. What Determines Case?
2. The Current Study
   - Method
   - Results
3. Discussion
4. Conclusion

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Section 1

What Determines Case?

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Meaning difference

(3) a. Susi reist in die Heimatländer der Schulkinder.
PR travels in the-ACC homelands of the school children

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accusative	:	"directional" meaning
dative	:	"locational" meaning

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Meaning difference

(3) Susi reist in die/den Heimatländer/n ...
PR travels in the-ACC/DAT homelands/-DAT ...

"Susi travels in(to) the homelands ... ."

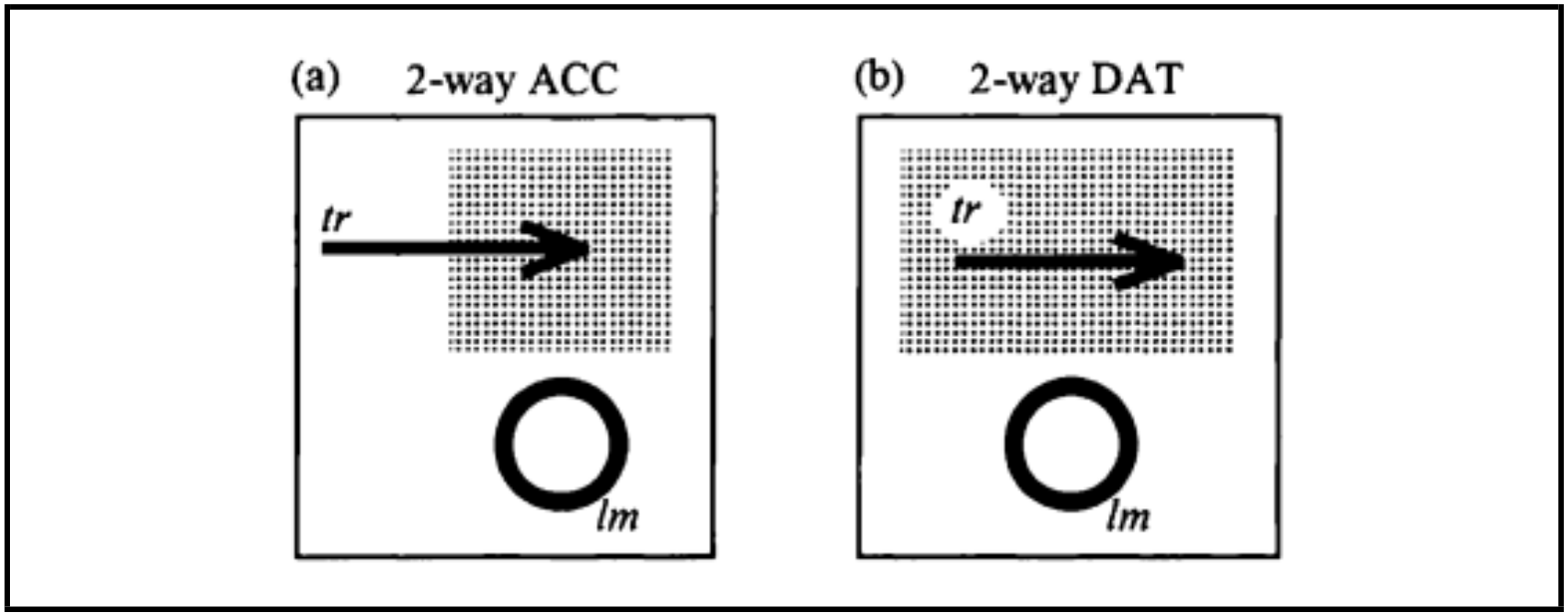

Figure: Trajector entering search domain (accusative) or not (dative) (Langacker, 1999, p. 35)

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Exceptions to the rule and influencing factors

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Exceptions to the rule and influencing factors

- other prepositions

(1) Najara ging zu dem alten Mann hinüber.
PR went to the-DAT old man over

"Najara went to the old man."

- multiword units with non-local meanings (Baten, 2009)

(4) Er verliebt sich in das Mädchen.
he falls in love REFL in the-ACC girl

"He falls in love with the girl.

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Exceptions to the rule and influencing factors

- Verb classes attract different cases (De Knop, 2008)

(5) Ich setze mich in meinen Wagen.
I sit down me in my-ACC car

"I sit down into my car.

(6) Ich sitze in einem Kästchen.
I sit in a-DAT box

"I sit in a box."

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- Other verb classes, syntactic role of PP, particle verbs, transitivity, voice, POS of lexical head, ...
(Rehbein and Genabith, 2006; Rys, Willems, and De Cuypere, 2014; Smith, 1995; Willems, 2011; Willems, De Cuypere, and Rys, 2018)

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Limits of previous studies

- Introspective
- Very small sample sizes
- Cover only handful of lexically specific contexts
- Little detail and explanation

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Section 2

The Current Study

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Subsection 1

Method

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Sample Preparation

- Leipzig Corpus Collection (Goldhahn, Eckart, and Quasthoff, 2012)
- 1-million-sentence corpus of mixed German
- POS-tagged and parsed with Zurich Dependency Parser (Sennrich, Volk, and Schneider, 2013)
- Sampled 1,000 sentences for each of nine two-way preposition

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Table: Distribution of prepositions in the sample

	an	auf	hinter	in	neben	über	unter	vor	zwischen	Total
n	1186	1083	970	1396	966	956	931	975	869	9332
p	12.71	11.61	10.39	14.96	10.35	10.24	9.98	10.45	9.31	100

Note n = number, p = proportion in %

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Coding and Statistical Analysis

Sample was coded for...

- Syntactic role of PP
- POS of lexical head and complement
- Particle verbs
- Voice
- Transitivity
- ...
- Individual prepositions
- Lexical heads and complements

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Coded sample was analyzed using ...

- Collostructional analysis (Gries and Stefanowitsch, 2004)
- Mixed-effects logistic regression (Johnson, 2008, pp. 216–258, Levshina, 2015, pp. 253–276)

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Subsection 2

Results

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Prepositions Attract Case

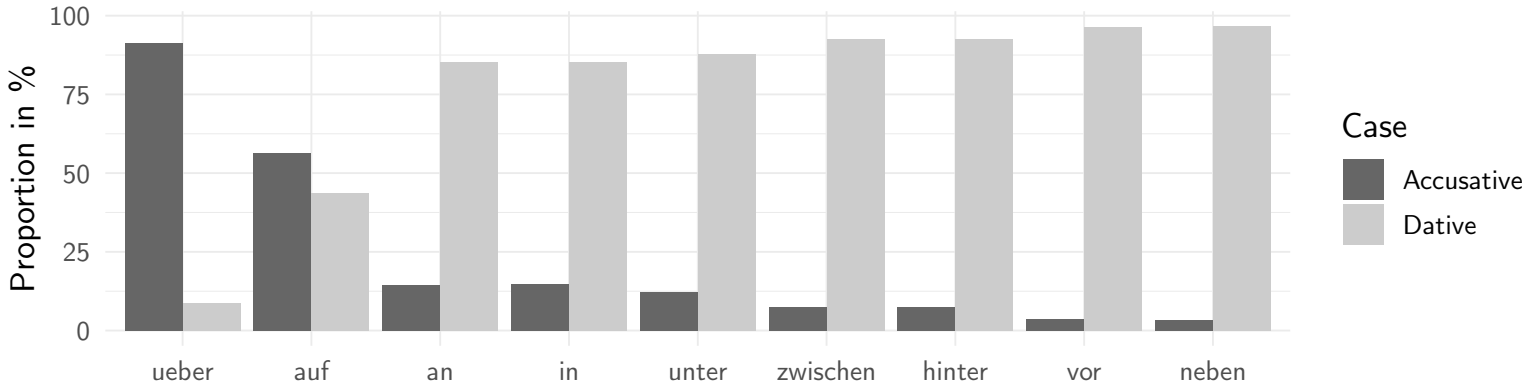

Figure: Distribution of prepositions across accusative and dative

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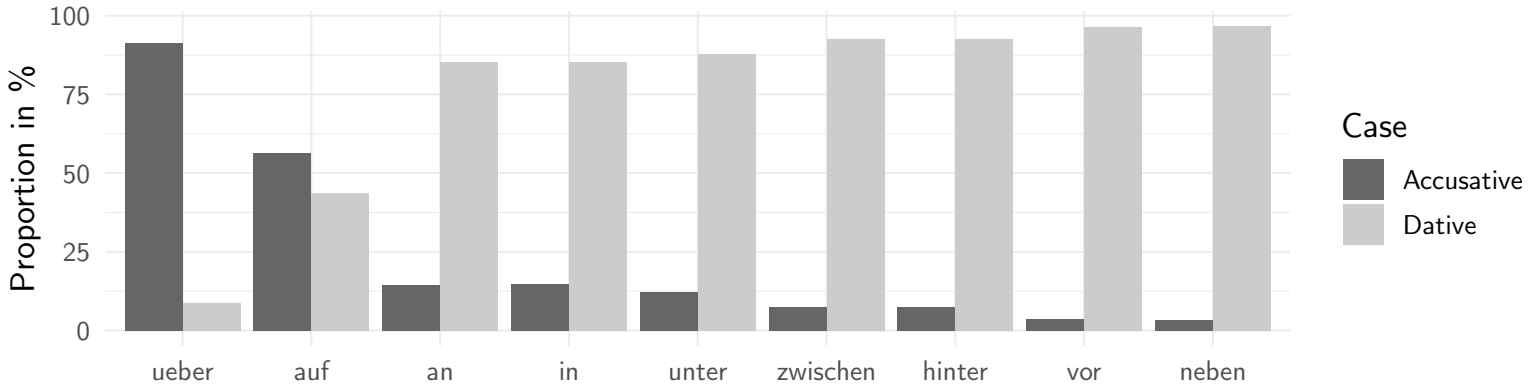

Figure: Distribution of prepositions across accusative and dative

Table: Results of collostructional analysis of case and individual prepositions

Preposition	über	auf	an	in	unter	zwischen	hinter	vor	neben
Preferred case	Accusative	Accusative	Dative	Dative	Dative	Dative	Dative	Dative	Dative
-log10(p)	Inf	139.14	15.13	17.64	18.47	37.57	41.93	71.76	72.69
ΔP	0.76	0.37	-0.1	-0.1	-0.12	-0.18	-0.18	-0.22	-0.22

Note: Statistical significance at -log10(p) >= 1.3.

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Lexical Heads Attract Case

... **V** Prep NP

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Lexical Heads Attract Case

... **V** Prep NP

Figure: Results of collostructional analysis of case and lexical heads (with all $p < 0.05$ and size indicating ΔP)

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Lexical Complements Attract Case

... V Prep **NP**

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Lexical Complements Attract Case

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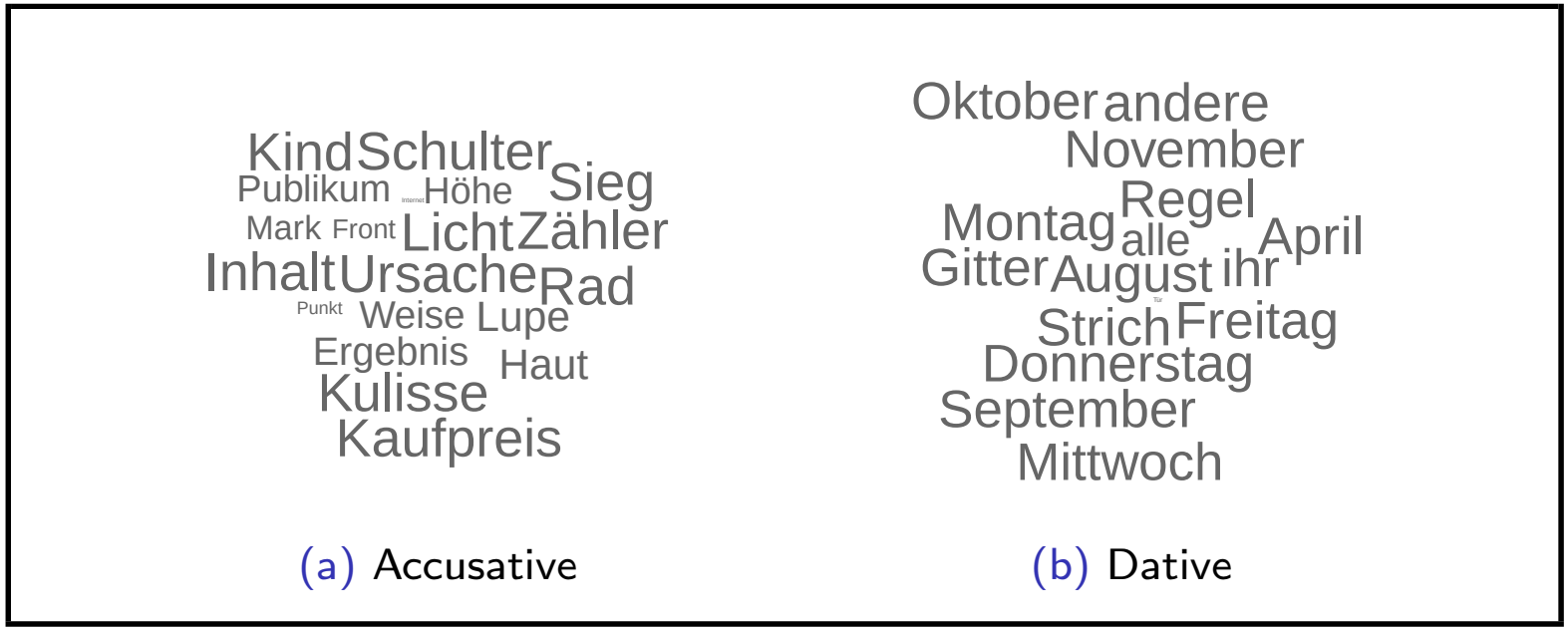

(a) Accusative (b) Dative

Figure: Results of collostructional analysis of case and lexical complements (with all $p < 0.05$ and size indicating ΔP)

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Little influence of other factors

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Little influence of other factors

Table: Results of logistic regression analysis predicting dative case

	b	*SE*	*z*	*p*
Intercept	2.15	0.62	3.46	***
Contracted (vs stand-alone) preposition	1.63	0.15	11.11	***
Transitive (vs intransitive) verb	-0.39	0.08	-4.86	***
Location	-0.79	0.05	-15.35	***
Past participle (vs no past participle)	-0.53	0.09	-5.83	***
Particle verb (vs no particle verb)	-0.31	0.15	-2.06	*
Adjunct (vs object)	1.17	0.08	14.3	***
Complement cue validity	0.1	0.01	7.37	***
Number of obervations	9332			
Kappa	8.48			
Predictive accuracy	0.88			
Predictive accuracy (no prepositions)	0.77			
Predictive accuracy (naive)	0.77			

Note $^{*}p \leq .05$; $^{**}p \leq .01$; $^{***}p \leq .001$.

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Model Estimates for Prepositions

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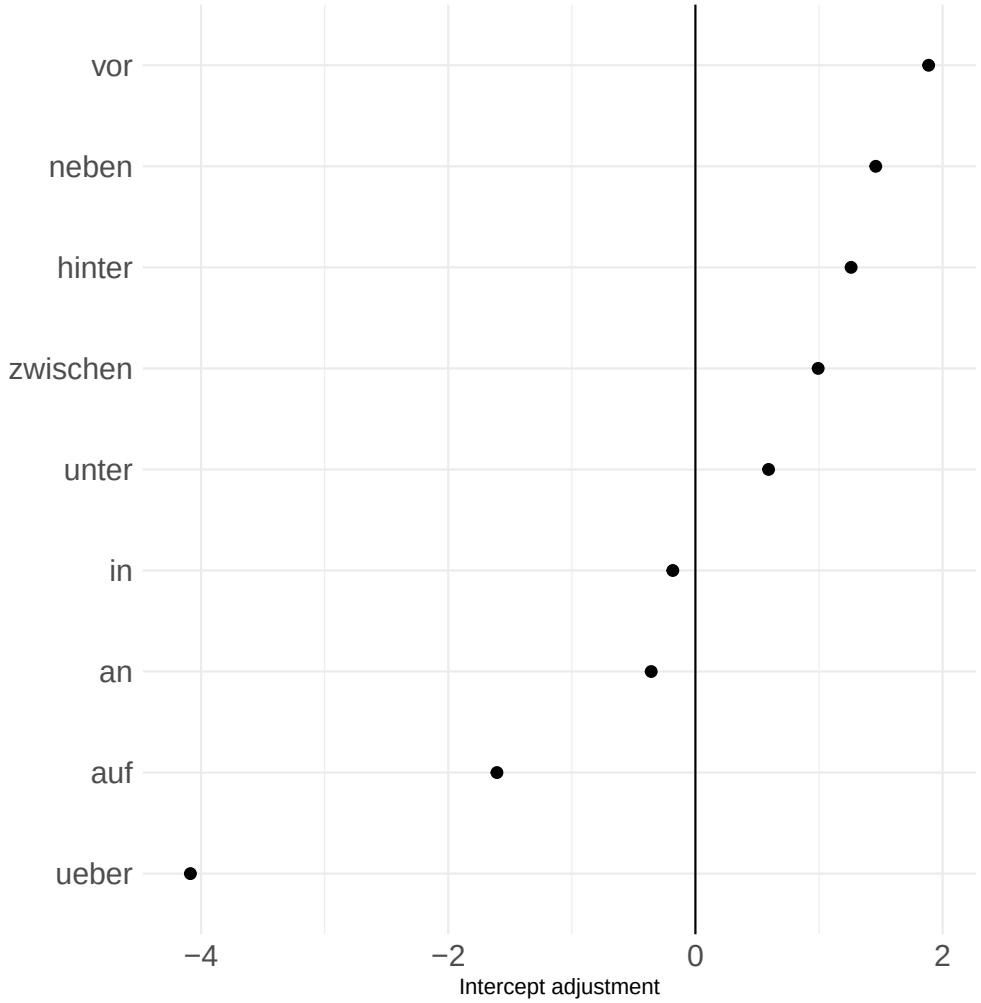

Figure: Intercept adjustments for individual prepositions in the regression model predicting dative case

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Usage-based Constructions

Results indicate that ...

- Case of two-way prepositions depended on lexical context
- Higher-level factors had little influence on case

... suggesting a **usage-based grammar network** (Diessel, 2019).

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- Connection strength depends on frequency of co-occurrence.

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Low-level Chunked Lexical Strings

Invariant lexical strings are chunked and stored at low levels of representation.

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denken an $\rightarrow$ $\text{NP}_{\textbf{Acc}}$
("think of")

Angst vor $\rightarrow$ $\text{NP}_{\textbf{Dat}}$
("fear of")

freuen über $\rightarrow$ $\text{NP}_{\textbf{Acc}}$
("be glad about")

warnen vor $\rightarrow$ $\text{NP}_{\textbf{Dat}}$
("warn about")

verzichten auf $\rightarrow$ $\text{NP}_{\textbf{Acc}}$
("relinquish")

sterben an $\rightarrow$ $\text{NP}_{\textbf{Dat}}$
("die from")

...

...

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Mid-level Prototypes

Accusative and dative attract different groups of meaning-related verbs and specific prepositions. Language users extract these distributional usage patterns from their input.

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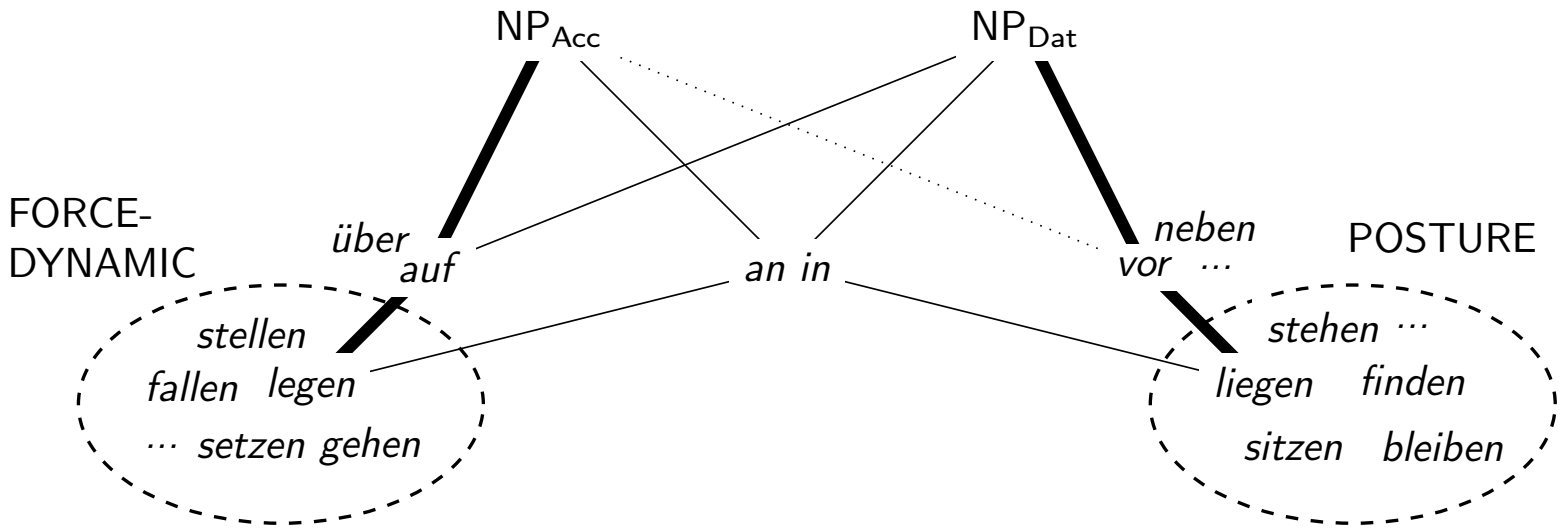

Figure: Network of case constructions and associated lexical items; line strength indicates strength of associative connection

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Prototypes Determine Case in Similar Contexts

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(7)

Mitten	im	Ort	befindet	sich	ein
in the middle	in-the-DAT	village	is located	REFL	a

Campingplatz.
camping site

"There is a camping site located in the middle of the village."

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the	company	steps	on	the-ACC	cost breaks

"The company cuts down costs."

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the	company	steps	on	the-ACC	cost breaks

"The company cuts down costs."

(9)

Ich	wartete	jetzt	jeden	Tag	auf	das	Ergebnis.
I	waited	now	every	day	on	the-ACC	results

"I waited for the results every day."

Ger	Eng
an	"at"
auf	"on"
hinter	"behind"
in	"in"
neben	"beside"
über	"above"
unter	"under"
vor	"before"
zwischen	"between"

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Productive Extensions to Case in Novel Contexts

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Productive Extensions to Case in Novel Contexts

(3) a.

Susi	reist	in	die	Heimatländer	der	Schulkinder.
PR	travels	in	the-ACC	homelands	of the	school children

"Susi travels into the homelands of the school children."

Ger	Eng
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Productive Extensions to Case in Novel Contexts

(3) a. Susi reist in die Heimatländer der Schulkinder.
PR travels in the-ACC homelands of the school children

"Susi travels into the homelands of the school children."

b. Susi reist in den Heimatländern der Schulkinder.
PR travels in the-DAT homelands-DAT of the school children

"Susi travels in the homelands of the school children."

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Section 4

Conclusion

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Conclusion

- Case of two-way prepositions depends on lexical context
- Lexicogrammatical constructions connect lexical items and case at different levels of schematicity
- Item-specific prototypes associate cases with groups of meaning-related verbs

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Two-way prepositions are no grammatical "misfits" anymore but part of network in which case is connected to prepositions and other lexical items depending on their use.

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Two-way prepositions are no grammatical "misfits" anymore but part of network in which case is connected to prepositions and other lexical items depending on their use.

Future studies should focus on effects of lexical strings on case and in general the sequential predictability of case from lexical context.

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Thank you for your attention!